

Maelstrom: Spatial Instrument and Decision Making Tool Based on Pattern Progression

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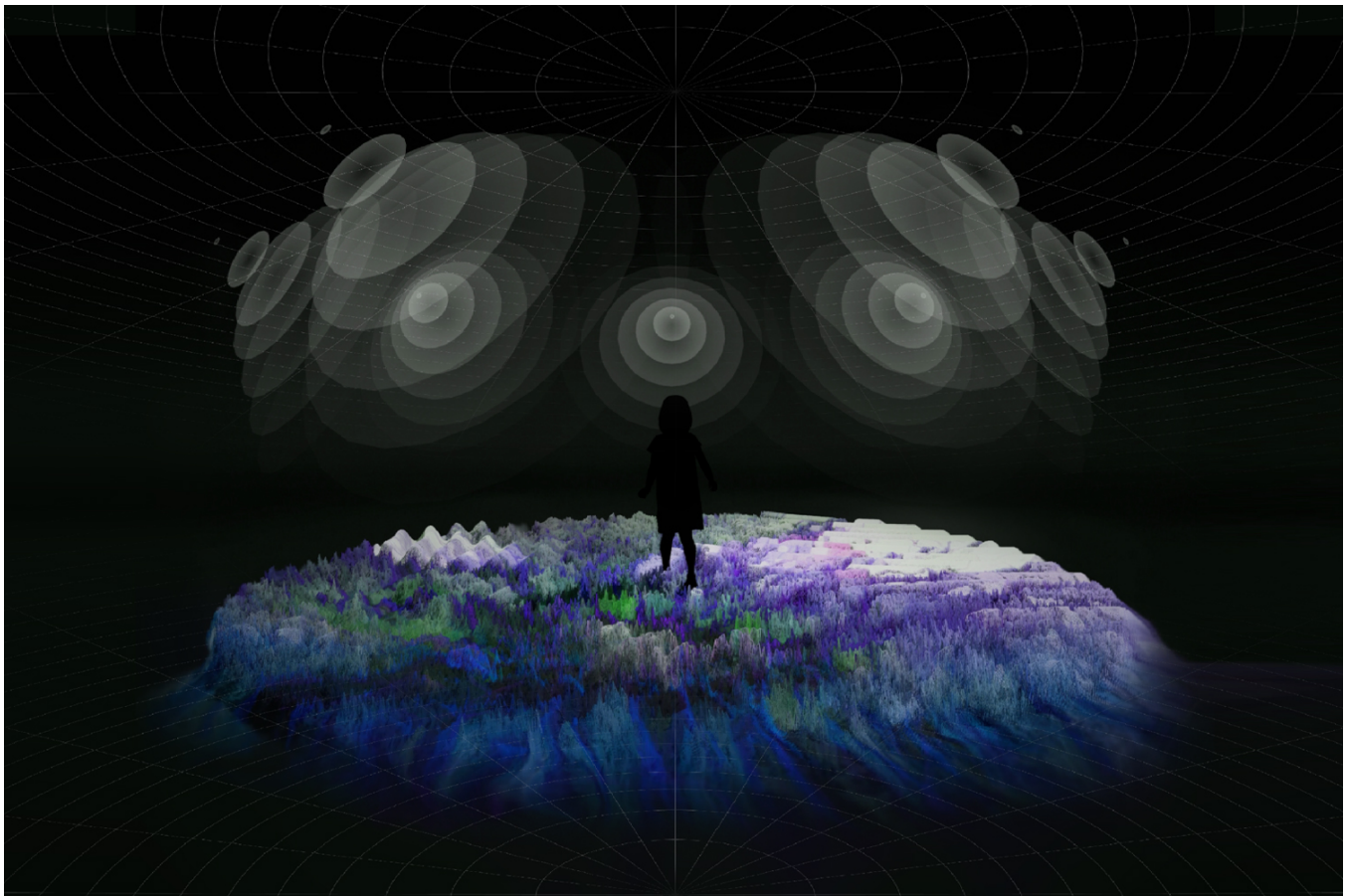


Figure 1: 7.1 channel audio guiding audience through the landscape of generative maps.

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ABSTRACT

Maelstrom is a spatial instrument that combines projection with interactive spatial audio, performed by the audience in the experience area. It is an installation that focuses on visual and acoustic pattern recognition, serving as a tool for individuals to navigate environments overwhelmed by messages. It provides audio-guided

experiences, allowing the audience to make decision based on different audio cues and explore visuals as maps developed by a generative model of artificial intelligence. This project attempts to challenge the sensory hierarchy and accessibility within immersive technologies, questioning the cognitive capabilities of the human brain.

KEYWORDS

Immersive Experience, Spatial Computing, Sonification, Spatial Audio, Interactive Music

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1 BACKGROUND

Within the domain of architectural studies, the concept of space is conventionally understood as an environment characterized by its material structures and static visual components. Dynamic elements and human-centered sensory experiences, though often proposed, frequently fail to materialize in final architectural products. The discipline predominantly gravitates towards the allure of drawings and visual forms, constrained by the static nature of materiality. This often leads to a disconnect between the envisioned dynamic experiences and their translation into tangible forms, overlooking the human perspective and the dilution of the original concept. In practice, while building accessibility and acoustic design are addressed through regulations, their potential for creative, non-visual spatial experiences remains largely untapped. This oversight marginalizes certain demographics, notably individuals with visual impairments. In book *"Spaces Speak, Are You Listening?: Experiencing Aural Architecture"* [2], the ways our brain perceiving visual and acoustic space functions differently, but the overall instructive mindset in the field stays ocularcentric. In a broader sense, the art world involves more mediums, providing seemingly more options for experiences. In recent years, the term "multisensory" is becoming abused, typically encompassing an assemblage of sensory elements arranged within a visually driven narrative, still entrenched in the linear and static ocular-centric culture. Discussions around terms like "soundscape" tend to be limited, focusing on spatial audio techniques such as simulating sound localization and environmental occlusion, or creating atmospheric audio using sound samples in a space. This treatment of "multi-sensory" elements, especially prevalent in media art, often fails to forge meaningful connections between interactive or non-interactive visuals and their accompanying soundscapes, seldom offering fresh cultural insights or innovative thought processes in sound design. This hierarchy of senses not only reinforces disparities in perception but also systematically excludes certain demographic groups that may interpret narratives through distinct experiential lenses.

The field of music and sound, influenced by ocular-centric perspectives, often relegates sound to a supplementary role in interactive designs, where narratives remain inaccessible without visual aids or graphical interfaces. The dominance of tactile and visual

interfaces in music, particularly since the introduction of hardware synthesizers, further exemplifies this trend. The advent of Digital Audio Workstations (DAWs) and graphical Digital Signal Processing (DSP) software, like Max/MSP [6], revolutionized the industry, yet they often require intricate knowledge or training for effective use. Notable systems like Iannis Xenakis's Upic [5] and the subsequent Iannix [4] explored graphic, geometric, and spatial controls for audio synthesis and manipulation, but they were primarily designed from a top-down approach for professional musicians, with little consideration for the listener's spatial experience in the sound field. This dynamic foster a binary division between performers, who interact directly with the narrative, and audiences, who are defined as a passive role with inability to direct the narrative.

In terms of scientific researches, sonification has been utilized to interpret data with sound while its subcategory Pulsed Melodic Affective Process (PMAP) [1] focuses more on emotional and affective aspects. Influenced by ocular-centrism, sonification has been defined as a supplemental method, while the aesthetics of sonification itself, remain underexplored and distant from public since the beginning – sounds are straight, plain, or minimal for mere scientific researches in most situations. Visual is prioritized – interfaces or artworks driven by such mindset presents strong ocularcentric tendencies [8], and acoustic-focused interfaces are underexplored. The methods of sonification are driven by the visual mindset which stress complexity but in fact creating chaos – complexity that disconnect with the meaning and characteristics of data, due to challenges on chaos caused by the richness of sonic information, which is widely observed in most mass data sonification [3]. The exploration of complexity, particularly through sonic experiments within certain artworks, often leads to outcomes and methodologies that warrant critical examination. This prompts a reevaluation of whether such principles intrinsically pertain to all sensory experiences or are a manifestation of visual culture's influence on our auditory interpretations. Against this backdrop, our project proposes an alternative approach, facilitating narrative engagement through a framework centered on acoustic-driven spatial interactions.

2 INTRODUCTION

From a historical overview of visualization, sonification, music, and architecture, this project embeds the latest interactive technologies to innovate human-centered narratives. These narratives are driven by spatial interaction and acoustics, enhancing accessibility, and providing affective feedback for general audiences, including those who are visually impaired. With a spatial musical system, the system assists navigating space of sensory overload, offering a critical perspective on prevailing ocularcentrism and sensory hierarchy. It provides a multisensory method to augment the decision-making process and navigate the current maelstrom of messages with a musical progression framework. 'Maelstrom' presents a journey through the symbiotic relationship between humans and the environment, with a groundbreaking real-time audio-visual interactive interface. This piece reflects on the prevailing overflow of data and technology that shapes our information ecosystem and the sensory overload imposed on humanity. It delves into the impact of "immersive" information environments and generative technologies on our

cognition: within today's data ecosystem, how do immersive experiences and generative technologies influence our capacity to process information? How can we perceive patterns in otherwise seemingly chaotic and unpredictable streams of activity as a means of not only surviving but also embracing the inevitable interdependence of an unpredictable future? Marshall McLuhan recognized parallels of today to Edgar Allan Poe's novel "A Descent into the Maelström"[9] as a comparative situation of current era of message[7]. In this short story, a sailor becomes so absorbed in thought that he is caught in a maelstrom. Realizing he cannot row his boat out of the situation, he must observe the objects within the maelstrom and grab onto the recurring ones as a means of escape. As an analogy to this vortex, the Maelstrom project embodies the challenges of our information age today, and the exhibit replicates the process of "detachment" through information overload, yet it possesses cyclical patterns in time. Through such sensory patterns, memories and habits become an inheritable feature of the environment, offering a way out of the maelstrom.



Figure 2: The experience of Maelstrom. Demo by Xiaoxiao Zhao. Photo by LeleTown.

Crafted with the Unreal Engine Metasounds and Dolby Atmos spatial audio, Maelstrom envelopes audiences in an spatial-visual and auditory experience through 7.1 surround sound and video projections. Similar to an instrument, the installation allows performance through spatial interaction with the interface. Guided by the console, visitors can manipulate the projections and perform the interactive spatial audio with a joystick controller, navigating through the created environment as if meandering amidst it. By tweaking the controller at the center, the audience control the movement of camera in virtual space, changing circular projection on the ground and rotating soundscape around the center, reflecting their relative position in the vortex of Maelstrom. The installation is an interactive piece controlled by one person, but shared by everyone in the projection area and sound field. Following auditory cues from beyond their visible range, they traverse a network reminiscent of satellite imagery and biological networks. As participants move outward from the vortex's core, the turbulence and noise wane, and direction becomes more discernible, ushering in the subsequent scene. These scenes, while analogous, bear their unique distinctions:

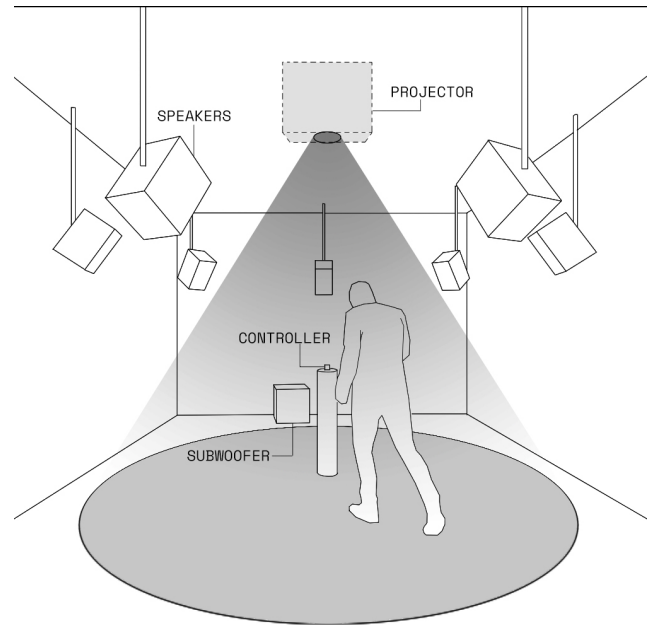


Figure 3: On-site detail of Maelstrom - 7.1 surround sound system and projection presents a multi-sensory narrative. Audience control the movement and rotation to navigate the virtual space.

visually, they resemble the shifting terrains of maps from different epochs, a portrayal of inherited, evolving, and rejuvenating images. Aurally, they resonate with electronic music and live sound performances composed of recursive sonic landscapes—within these cyclical transitions, participants navigate the space swiftly, drawing upon their past habits and memories. The audience and “Maelstrom” gradually shape and adapt to each other, embarking on a journey about memory and adaptability amidst an environment teeming with intricate information.

The journey from the center of the circular vortex to the outer edge presents a transition from a visually guided narrative to an audio-guided one. At the beginning of each cycle, when a new scene is revealed, the audience is placed at the center of the vortex, where the scene is more visible, while sounds are less directive and more noisy. As the audience approaches the edge, where spatial audio objects are located, visuals fade away and audio cues become more distinctive, guiding the audience to navigate the darkness and make decisions between different sound objects from all directions that directly link to the next phase of visual and musical progression. In our early experiments, we observed challenges in the users' ability to deal with multiple audio cues in space – the concurrency of sound, frequency distribution of different audio sources, and various audio effects – which introduced chaos and disturbance. In our final installation, such chaotic situations are intentionally designed for each scene, where from the center to the edge, users experience a decrease in information richness and are forced to rely on audio cues to escape the chaotic scene, finally finding their way “out of” the maelstrom/message vortex.

3 ACOUSTIC NARRATIVE

Auditorily, "Maelstrom" is an interactive sound installation that prompts audiences to learn and react to different messages in the environment, further understanding their relationship to surroundings through an evolving sonic narrative. In this narrative, audio cues movement, and audience's movement redefines the audio, delivering a multisensory experience. With our spatial music structure, which is centered on the method of Distance-related Audio Effect Modulation and directional audio spatialization, the project transcends traditional audio-visual experiences and becomes an interactive musical instrument where the audience can engage performance through a minimal interface. It redefines musical conventions with spatial and non-linear narratives, where music progression unfolds into different branches based on different target objects reached by the audience, advancing beyond mere sonification and visualization. Guided by spatial audio, the installation empowers all participants, including those who are visually impaired, to partake as independent performers. The narrative is crafted as an accessible learning journey, inclusive and shared by all, not exclusively reserved for professional musicians. Technically, with the latest technologies from Unreal Engine, Metasounds, and a distributed speaker array, it extends the interactivity of Dolby Atmos spatial audio.

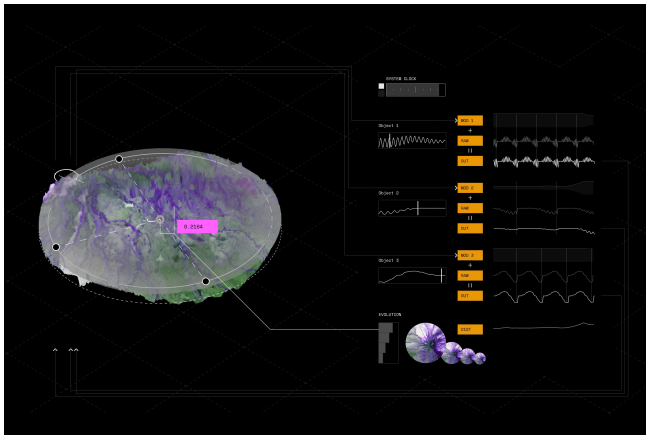


Figure 4: Framework of spatial music structure. In our case, three target audio objects are placed at the edge, guiding users to move through the maps of generative images.

Inspired by DJ performances, as well as music production in electronic music genres (e.g. techno), where musical progressions in certain situations are realized through audio effect modulation over repetitive patterns (e.g., the increasing frequency parameter of a low-pass filter on a looping synthesizer sequence) - we have developed a Distance-related Audio Effect Modulation. This approach translates the relative distances between a user and target audio objects into audio effect modulation parameters, forming a responsive audio narrative in space. By linking distance to such modulation, the narrative informs the user of changing data while simultaneously producing musical feedback as they move through space. Unlike traditional sonification methods that transform data into non-musical or non-affective parameters such as volume and pitch,

our method emphasizes affective feedback and musical possibilities through interaction—translating data into parameters like the frequency of a filter, the grain size of granular synthesis, the amount of overdrive, and the delay time. In Maelstrom, we envision that users can learn about their relationship to their surroundings by distinguishing spatialized audio objects from the composed acoustic message, gradually understanding how music responds to their movement and how they can develop musical progressions. These progressions are not linear, especially when multiple audio objects guide the audience to make decisions by walking in space—each direction and each target audio object represent the direction in which the music is developing, which could serve as 'multi-branch music.' This musical structure evolves into a recursive tree-like structure that links one pattern to multiple possibilities, offering a human-centered perspective on musical performance and allowing audiences to participate as performers.

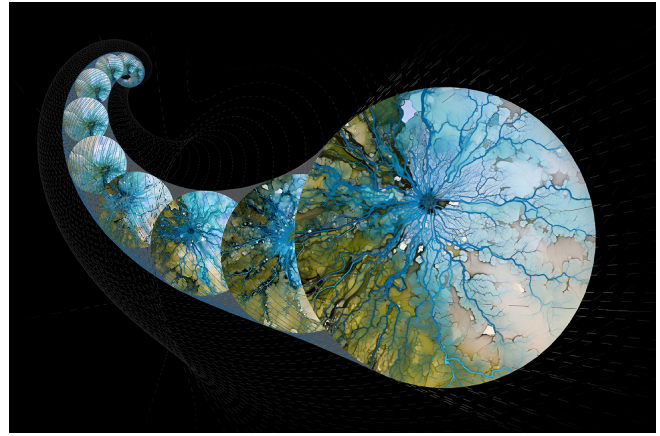


Figure 5: The original image of each scene is partially removed and regenerated with the DALL-E 2 Inpainting function. Throughout iterations, the map gradually evolves into a different pattern.

4 VISUAL NARRATIVE

The visual component of "Maelstrom" is powered by a generative AI model, which enables user-directed variation in the imagery. This feature positions the audience at the heart of an evolving landscape, seamlessly transitioning from one scene to the next. Utilizing DALL-E 2 and Midjourney generative models, the installation crafts maps that echo urban satellite imagery, mirroring processes akin to geological shifts and generational changes. It offers users a navigable map, enabling them to traverse and adjust to new landscapes based on their recollection of previous maps. This continuity, echoing characteristics and memories passed down through generations, aids the audience in adapting to novel environments by drawing on their accumulated experiences. The transition between scenes, coupled with the sensory memory of their attributes, builds a repository of experience that informs the audience's movement within the space. Concurrently, the audience's decisions, informed by auditory signals, influence the scene's evolutionary trajectory, fostering a dynamic interaction with the environment. With each

cycle, as a new scene unfolds, the preceding map is partially erased and overlaid with new patterns, recentering the audience within this transformative space. These evolving patterns emulate a liquid vortex, generating visual feedback that enhances the immersive experience.

In the installation, the visuals are limited to a circular area of projection on the ground, where sound becomes more reliable and distinctive than the range of visuals. When users move around in the virtual space with the joystick, the projection presents a top view of the map, placing users inside the vortex of messages.

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